

ting from their 'intelligent virtual assistant.'

Before implementing Watson to regular use, a pilot project of six months was undertaken to test its worthiness for the treatment. Of the 200 test cases, only six per cent were found to be non-conforming, that is, the cases in which doctors' recommendation was different from Watson's.

"Watson continuously learns from the information shared on cases coming to the hospital—both related to data and doctors' consultation. It has an amazing artificial conscience," shares Venitha.

As a patient's case is brought to a doctor, while Watson analyses the patient's data and case history, the doctor provides the consultation and moral support. The final treatment is started with recommendations from both the doctor and Watson collaboratively.

The biggest concerns are about the safety of patients' information shared with IBM's platform and validity of treatments recommended.

While the government is yet to come up with a policy on safety, Manipal Hospitals maintains that they are using IBM's supercomputing and cloud services for analyses. The patient's record is owned by the hospital itself.

IBM Watson has been implemented as a monthly service structured and supported by the IBM cloud. To update Watson, two IBM employees have been stationed at the hospital. The maintenance of Watson is part of the contract and taken care of by IBM.

Artificial intelligence technology is all set to transform healthcare. IBM is one of the leaders, with most advanced system capabilities. Also, having a counter-intuitive resource base in the US brings about greater trust. Not just IBM but Manipal Hospitals is also seeking to expand the use

of technology directly to patients across geographies, in order to enable better health care. **EFY**

—Nidhi Arora, executive editor, **EFY**

Business Opportunities For DNA Testing Kits

➤ **Cost to start the business is relatively low and return margins are huge**

The market for genome data testing is disrupting globally. The direct-to-consumer genome testing market is expected to be worth 340 million USD by 2020, growing annually at 25 per cent—according to the latest findings by Credence Research.

DNA testing kits are used to conduct all kinds of DNA tests on plants or humans. The demand for these kits has increased manifold in the recent decade due to commercialisation of DNA testing for humans and use of the results for various health and lifestyle benefits.

Dr Saleem Mohammed of XCode (DNA testing kits provider for both consumer and clinical genomic applications) shares, "India is increasingly becoming a genomics hub due to low-cost testing as well as availability of skilled IT resources. Core investments for DNA testing kits business go into people, R&D on machine learning, AI and automation."

India is playing a significant role in the growth of DNA testing kits business globally. Various companies in the west (America, Canada and Germany primarily) are looking for partners to offer various kinds of DNA kits in India. With companies like Qaigen (Germany-headquartered DNA kits provider) setting up offices in India, manufacturing in the segment is also slowly picking up.

Dipender Singh of Royal Life Sciences, Indian partner for MIDi Sherlock USA, a company with clients across the globe, informed that the demand of their products had increased about ten times in India in the last seven years. Almost 30 per cent of their clients are from other countries as well.

Conventionally, DNA testing kits were used by labs and research centres to diagnose the DNA sequence for research and forensic purposes. However, with 23andMe (funded by Google) and Ancestry (offshoot of National Geographic) leading the R&D of DNA testing, companies around the world are working towards making DNA testing kits available to the masses. The US Food and Drug Administration (USFDA) has approved DNA tests by 23andMe to predict the risk of developing Parkinson's disease. With such strong organisations in the front, DNA testing kits is today one of the most lucrative business opportunities for investment.

Direct-to-consumer DNA testing kits have become common, subject to the development and investment in the business of bio-technology. Commercialisation of genome testing has attracted the attention of a lot of enterprises, research institutions and government bodies to the segment. Some of the market players in India include DNA Labs India, Map My Genome, Thermo Fisher, MedGenome, XCode and Indian Biosciences.